OLISEMEKA OKPALEKE

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https://emeka-portfolio.vercel.app/ | Github: https://github.com/Emeka-Okpaleke

PROFESSIONAL SUMMARY

FullStack Developer and AI Systems Engineer with 4+ years of experience building scalable web and mobile applications, backend APIs, and production ready AI driven systems. Proficient in Python, FastAPI, Node.js, SQL, React/React Native, and AI/ML workflows. Experienced in developing and specialized in architecting end to end LLM pipelines, semantic search infrastructure, integrating data analytics, and delivering seamless user experiences.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, HTML5, CSS3, Python
Frameworks & Libraries: FastAPI, Node.js, Express, React.js, Next.js, React Native, Tailwind CSS, Three.js
Databases & Data Tools: MongoDB, PostgreSQL, NumPy, dbt, Numpy
AI / ML Tools: Ollama, Vector embeddings, RAG, YOLOv8, OpenCV, LLM orchestration, Cosine similarity
Version Control & DevOps: Git, GitHub, CI/CD, Vercel
Testing & Performance: Jest, React Query, Redux, Zustand, Responsive Design

WORK EXPERIENCE

Jan 2026 – Present **AI Systems Engineer – EnvisionIt**

- Engineered a semantic deduplication system using transformer-based embeddings (Ollama + Llama3.2), reducing duplicate story creation by ~85% and improving content consolidation accuracy across continuous news ingestion cycles.
- Architected a RAG-style retrieval pipeline to inject relevant prior context during story rewrites, improving narrative coherence and reducing factual inconsistencies in AI-generated updates.
- Implemented high-dimensional (3072-d) cosine similarity matching for real-time article clustering, decreasing manual review workload by approximately 70% and enabling near-instant detection of related evolving stories.
- Built event-driven inference architecture that activates generative models only when similarity thresholds are exceeded (>85%), reducing unnecessary GPU runtime by over 65% and significantly lowering infrastructure burn.
- Deployed system components on AWS with scalable compute design, enabling automated scraping every 10 minutes while maintaining predictable resource utilization.

Aug 2023– Jan 2026 **Software Engineer | Truparse Nigeria**

- Built and scaled responsive web and mobile applications for short-term rental and e-commerce platforms using React, Next.js, and React Native.
- Developed backend APIs and microservices using Node.js and Python FastAPI, handling real-time data and analytics pipelines.
- Optimized data handling with PostgreSQL and MongoDB, improving API response times by 40%.
- Collaborated in Agile teams, participating in code reviews, debugging, and performance audits.

July 2025 – Dec 2026 **Full Stack Developer(Technical Lead) | Skills2Evolve (S2E) (Contract)**

- Led the development of the company's Learning Management System (LMS) using Next.js, MongoDB, and Prisma, deployed on Vercel.
- Built secure role-based dashboards (Student, Admin, Accountant, Publisher) with full authentication and authorization
- Integrated Sanity CMS, Resend, and Google Analytics to support content workflows, transactional emails, and analytics tracking.

- Implemented GDPR-compliant cookie consent flows and analytics configuration.
- Managed CI/CD pipelines using GitHub → Vercel and oversaw multi-environment deployments

Mar 2025 - June 2025 **Mobile Application Developer | Stardelite LTD (Contract)**

- Developed and maintained the frontend of cross-platform mobile applications using React Native, ensuring smooth performance and responsive UI across iOS and Android devices.
- Successfully contributed to the launch of 3 live mobile applications, actively involved from development to deployment stages
- Collaborated closely with designers, product managers, and backend developers to transform Figma designs into pixel-perfect, user-friendly mobile interfaces.
- Participated in code reviews, sprint planning, and team discussions, promoting best practices and knowledge sharing within the team.

Feb 2022 - Mar 2023 **Software Engineer | Figur Africa**

- Built interactive financial dashboards using React and TypeScript to display real-time investment and transaction data.
- Improved usability and responsiveness across mobile and desktop environments.
- Participated in regular code reviews, contributing to improved code quality and maintainability through shared standards.
- Engaged in Agile sprints, stand-ups, and retrospectives, helping the team deliver features on schedule and continuously improve development workflows.

EDUCATION

B.Eng. in Electrical Engineering (Second Class Upper Division)

Benson Idahosa University | CGPA: 4.35/5.0

PROJECTS

- ***RAG Knowledge Base (Python, FastAPI, ChromaDB, Ollama / Groq, Pydantic, Next.js 14, TailwindCSS)*:** https://rag-kb-seven.vercel.app/
  Enterprise-grade knowledge base with **Retrieval-Augmented Generation (RAG)** powered Q&A. Upload documents (PDF, DOCX, TXT, Markdown), ask natural language questions, and get AI-generated answers with source citations.

- **Video Intelligence API *(Python, FastAPI, YOLOv8, PostgreSQL)*:** https://github.com/Emeka-Okpaleke/video-intelligence-api
  Developed an AI-powered API for automated video analysis, detecting people, drawing bounding boxes, generating annotated previews, and providing analytics endpoints. Designed for scalability and production-ready deployment with PostgreSQL backend storage.

- **HOTEL PRO:** https://hotel-pro-eosin.vercel.app/
  HotelPro is a comprehensive hotel management dashboard designed to streamline operations for hospitality businesses. The dashboard provides a centralized interface for managing various aspects of hotel operations including reservations, room management, billing, guest feedback, and room service.

CERTIFICATION

- **July 2023** Google Digital Skills for Africa
- **Jan 2025** National Youth Service Corps Certification